

Graphene Oxide Aerogel Beads Filled with Phase Change Material for Latent Heat Storage and Release

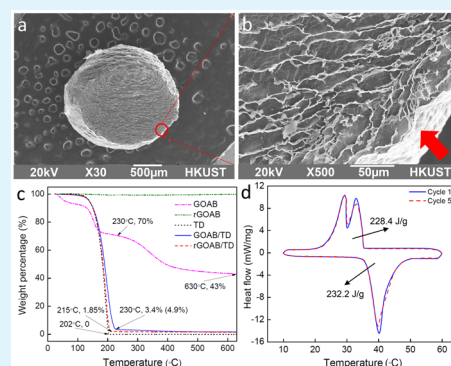
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Supporting Information

ABSTRACT: Phase change composite materials (PCCMs) with large latent heat capacity, a stable structure, and efficient thermal response have high potential for thermal management in various applications. Herein, novel reduced graphene oxide aerogel beads (rGOABs) are synthesized by wet spinning of GO slurry followed by thermal reduction, which are infiltrated with 1-tetradecanol (TD) paraffin as phase change materials (PCMs) to produce rGOAB/TD composites. An exceptionally high mass fraction, 98.8%, of paraffin TD encapsulated in the rGOABs is achieved in this study, which is known to be the highest among studies ever reported. It is demonstrated that the rGOAB/TD composite possesses a high latent heat value of 230.3 J g⁻¹ and maintains 96.6% efficiency after 50 heating–cooling cycles, making the composite PCM suitable for emerging thermal management applications. The thermal responsive tests of various samples indicate better thermal response of rGOAB/TD than GOAB/TD without thermal reduction or the expanded graphite counterpart. With a high thermal storage capability, a high heat transfer property, and high flexibility, the novel rGOAB/TD PCCM has great potential in thermal management applications, such as lithium-ion battery packs.

KEYWORDS: graphene aerogel, phase change material, thermal management, energy storage, lithium-ion battery pack



1. INTRODUCTION

Phase change materials (PCMs) have drawn extensive attention as one of the most promising candidates for thermal energy storage and management. With their high latent heats, excellent chemical compatibility, thermal stability, and the wide range of phase change temperatures, PCMs show great potentials for many different applications. For example, the reduction in cooling load of a building envelope in Singapore integrated with PCMs has been investigated.^{1,2} PCM-based thermal management for electric vehicles has been widely studied.^{3–6} However, low thermal conductivities, instability, limited volumetric energy storage densities, and flammability are prominent weaknesses of organic PCMs. Among well-studied PCMs, fatty acids and alcohols show advantages of relatively high enthalpies and low costs. 1-Tetradecanol (TD), as one of the fatty alcohols, with a high latent heat of over 220 J g⁻¹ and liquid–solid phase transition temperature of around 38 °C, is highly promising for tailored thermal management of lithium-ion batteries, the optimal temperature range of which is from 20 to 35 °C. In the meantime, a few strategies aiming at mitigating undesirable features of paraffins have been developed by modifying the properties of wax and the storage unit, such as the introduction of thermally conductive fillers, microencapsulation, and shape stabilization.⁷ Many different materials have been explored to increase the thermal conductivity while maintaining a stable structure during the solid–liquid phase transition, such as expanded graphite

(EG),^{8–19} graphite,^{20–24} carbon fibers,^{25,26} carbon nanotubes,²⁷ graphene,^{28–30} metal foams,³¹ molten salts,¹⁸ and porous organic polymers (POPs).^{32–35} Octadecane is encapsulated with a metallic–polymeric TiO₂–PUA shell to enhance the thermal responsive property where the core material fraction was only 73 wt % with a relatively high density.³² Low density carbon materials with good thermal conductivity have attracted much attention, among which EG has been most extensively explored. EG was introduced into a shape-stabilized PCM to maintain 80% of the pure PCM latent heat at 3–5 times the thermal response speed.⁸ A stearic acid–acetanilide eutectic mixture/EG composite PCM is reported to show a high latent heat of 176.2 J g⁻¹ with a thermal conductivity 14.5 times higher than that of pure PCM.¹² By compacting EG into different densities, paraffin wax/EG composites are reported to obtain thermal conductivity 28–180 times that of pure paraffin wax.¹⁶ An inorganic PCM, calcium chloride hexahydrate, was also encapsulated by EG to achieve 14 times the thermal conductivity, but the concentration of PCM only reaches 50 wt % to remain structurally stable.¹⁷ A minimum EG mass fraction of 10% was necessary for the PCM/EG composite to be structurally stable.¹⁴ Most directly applied PCM/EG composites are bulk materials with

Received: February 22, 2019

Accepted: May 1, 2019

Published: May 1, 2019

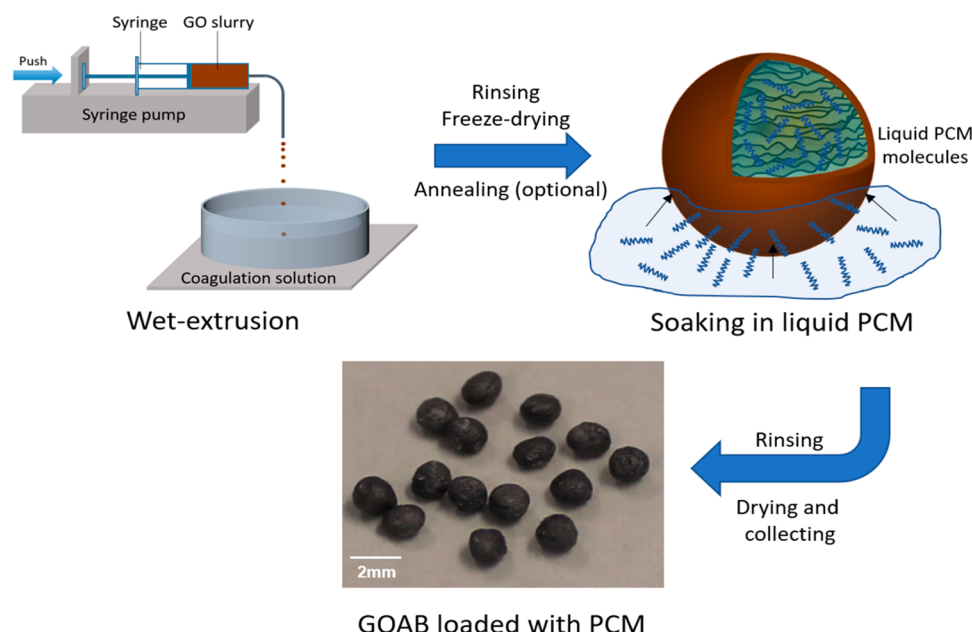


Figure 1. Schematic illustration of fabrication of rGOABs impregnated with PCM.

their mechanical properties scarcely reported, which however are an important safety factor for power battery packs. In addition, the electrical properties were hardly mentioned, even though a high electrical conductivity may have a negative influence on the performance of batteries through short circuit. Ceramic silicone rubber is reported to have an extraordinary mechanical strength, high thermal conductivity, and electrical insulating property, making it an ideal sealing matrix for power battery packs.^{36–43}

In this study, we present a facile method to fabricate a novel PCCM based on reduced graphene oxide aerogel beads/tetradecanol (rGOAB/TD), with up to 98.8% mass fraction of PCM, the highest ever reported, and a latent heat of 230.3 J g^{-1} which is reversible and stable after 50 heating–cooling cycles. Moreover, the rGOAB/TD composite shows even better thermal response than expanded graphite with 1-tetradecanol (EG/TD) and GOAB/TD after adding into a silicone rubber matrix. The GOAB/TD composite is more flexible than those previously reported bulky PCM composites, and it can be applied directly to an appropriate matrix to prepare electrically insulating and thermally conductive composites with good mechanical strength for practical applications.

2. EXPERIMENTAL SECTION

2.1. Materials. Chemically pure paraffin, 1-tetradecanol (TD, melting point $T_m = 38^\circ\text{C}$), was supplied by Shanghai Aladdin Biochem Technology Co. GO aqueous slurry (solid concentration: 1.5–2 wt %; lateral diameter: $0.2\text{--}10 \mu\text{m}$ and $5\text{--}40 \mu\text{m}$; single layer rate: >99%) was supplied by Shanghai Ashine Technology Development Co. Polyethylenimine (PEI with average molecular weight of ~ 800) was supplied by Sigma-Aldrich China, Inc. Expanded graphite (EG) with an expansion ratio of 300:1 was purchased from Qingdao Graphite Co. Polydimethylsiloxane (PDMS) SYLGARD 184 was supplied by Dow Chemical Co. Silicone rubber (SR) with thermal conductivity of 0.4 W/(m K) was supplied by Shenzhen Union Tenda Technology Co.

2.2. Preparation of RGOAB Filled with Paraffin and PDMS Composites. GOABs were fabricated via wet-spinning as reported previously.⁴⁴ The GO aqueous slurry was extruded through a syringe

needle into 1 wt % PEI aqueous solution to form a stable bead structure which was kept for 3 h before rinsing with deionized (DI) water. The collected wet beads were freeze-dried to obtain GOABs. The GOABs were subjected to thermal annealing at 1100°C for 1 h for reduction to obtain rGOABs. The solid-state paraffin TD was melted at 55°C in an oven, and rGOABs were soaked in the liquid paraffin for 5 min to collect the rGOABs impregnated with paraffin. The products were washed with ethanol at room temperature to remove the excessive paraffin on the surface of the beads, followed by air drying. The above synthesis process is schematically illustrated in Figure 1.

Different groups of GOABs and rGOABs were fabricated to investigate the effect of concentration, graphene sheet size, and thermal reduction, as shown in Table S1. S represents the GO slurry with small lateral sizes ($0.2\text{--}10 \mu\text{m}$ in diameter), while L represents that with large lateral sizes in the range $5\text{--}40 \mu\text{m}$. There is only a narrow range of GO slurry concentrations for beads to form during wet-spinning, and the numbers in denotations respectively represent the minimum and maximum concentrations for each group. “r” stands for the reduced samples. Table 1 shows important physical properties of GOAB and rGOAB prepared from a GO slurry with different concentrations and GO lateral sizes. It is interesting to note that the diameters of both GOABs and rGOABs only had a narrow range of variation. In terms of weight, density, and absorbability, the rGOABs had lower weight over GOABs, which is due to the removal of oxygenated functional groups on graphene oxide sheets, which leads to lower densities, given that the sizes are comparable. A lower density and comparable size of rGOABs contributed to the relatively high absorbability of PCM in terms of mass fraction compared with GOABs.

EG was compressed in a mold at a pressure of 0.35 MPa followed by soaking in liquid-state PCM to obtain an EG/TD composite with 90% mass fraction of TD and a total density of 0.85 g cm^{-3} . The GOAB/TD, rGOAB/TD, and EG/TD composites containing 1 g of TD and pure TD were dispersed uniformly in the PDMS matrix, which were treated by vacuum overnight followed by heating at 35°C for 10 h. Five different disk-shaped PDMS composites of 30 mm in diameter and 7 mm in thickness were fabricated for thermal responsive characterization.

2.3. Characterization and Measurements. The diameters and weights of GOABs and rGOABs before and after infiltration of TD paraffin were measured. The absorbability of GOABs and rGOABs was investigated by two methods, i.e., (1) thermogravimetric analysis

Table 1. Parametric Study of Weight, Size, Density, and Absorbability of GOABs and RGOABs Prepared from Raw Materials with Different Lateral Sizes and Concentrations^a

GOAB	av weight (mg/bead)	av size (mm)	density ^b (mg cm ⁻³)	absorbability (wt %)
S13	0.16 ± 0.02	2.5 ± 0.3	2.33	96.99
rS13	0.06 ± 0.01	2.3 ± 0.2	1.22	98.72
S15	0.18 ± 0.01	2.5 ± 0.2	2.65	96.84
rS15	0.07 ± 0.02	2.4 ± 0.1	1.21	98.40
L11.7	0.15 ± 0.03	2.3 ± 0.2	2.84	97.38
rL11.7	0.05 ± 0.01	2.3 ± 0.1	0.91	98.83
L13.5	0.17 ± 0.01	2.3 ± 0.1	3.15	96.92
rL13.5	0.05 ± 0.01	2.3 ± 0.1	0.91	98.77

^aS represents the GO slurry with small lateral sizes (0.2–10 μm in diameter), while L represents that with large lateral sizes in the range 5–40 μm, with r standing for the reduced samples. The numbers respectively represent the minimum and maximum concentrations for each group. ^bGiven that GOABs and rGOABs are not exactly spherical, here we take them as regular spheres and calculate their approximate density using their average weight and size. The real density of the beads should be slightly lower than the approximate one.

(TGA TA-Q5000) of pure TD paraffin, GOABs, and rGOABs impregnated with TD paraffin were heated to 800 °C at a heating rate of 20 °C/min in a nitrogen atmosphere and (2) calculation of the data from weighing rGOABs before and after infiltration with TD on an analytical balance based on the followed equation:

$$\text{absorbability} = \frac{m_{\text{rGOAB/TD}} - m_{\text{rGOAB}}}{m_{\text{rGOAB/TD}}} \times 100\% \quad (1)$$

where the masses of rGOAB/TD and rGOAB are denoted by $m_{\text{rGOAB/TD}}$ and m_{rGOAB} respectively, obtained by weighing the rGOAB before and after infiltration with TD on an analytical balance.

The morphology of the neat rGOAB and rGOAB/TD composites was examined by a scanning electron microscope (JEOL-6390). The composition of the composite was characterized by FT-IR spectra analyzer (Vertex 70 Hyperion 1000, Bruker) from 4000 to 400 cm⁻¹.

The thermal dynamic properties of pure TD, GOAB/TD, and rGOAB/TD composites during heating and cooling cycles were investigated by differential scanning calorimetry (DSC TA-Q2000). Thermal responsive tests were conducted by heating pure PDMS, EG/TD, GOAB/TD, and rGOAB/TD in PDMS on a hot plate, and their temperatures were monitored by an IR image camera (Compact

XR Seek Thermal). The thermal conductivities of different materials were measured on a thermal conductivity meter (Hot Disk TPS2500s). The compression tests were conducted on a Sintech 10/D Universal Testing Machine with furnace for temperature control.

3. RESULTS AND DISCUSSION

3.1. Morphology of RGOAB Impregnated with Paraffin. Figure 2 presents optical and SEM images of porous and filled rGOABs. The unfilled rGOABs exhibit a highly porous structure with interconnected graphene sheets. The liquid paraffin was absorbed through the cellular network of rGOABs by capillary force to form rGOAB/TD composites (Figure 2e–g). The absorption process was fast (<5 min), and the rGOAB was fully impregnated due to the large amount of interconnected microsized pores which provided the capillary forces and the pathway for paraffin to infiltrate. In the meantime, the interconnected highly porous rGO sheets provide amounts of pathway for heat to penetrate into PCM, which is beneficial to the thermal response of the material. The rGOAB surface shown in Figure 2d was smooth and dense membrane formed by coagulation of GO and PEI with no porosity observed at a magnification of ×10000, which prevented leakage of TD, holding the PCCM form stable. As shown in Figure 2f,h, the morphology of rGOAB/TD cross section showed little difference after 50 cycles of heating and cooling, which indicated the good cyclic stability of the composite.

3.2. Composition, Absorbability, and Thermal Properties of RGOAB Impregnated with Paraffin TD. The composition of rGOAB/TD composites was evaluated by FT-IR. The FT-IR spectra of 1-tetradecanol, neat rGOAB, and rGOAB/TD composite are displayed in Figure 3a. The FT-IR spectrum of TD presented prominent peaks at 3320 and 3210 cm⁻¹, both of which are attributed to the stretching vibration of the –OH group. The peaks at 2966 cm⁻¹ correspond to the stretching vibration of –CH₃, and 2932 and 2850 cm⁻¹ are ascribed to the stretching vibration of –CH₂–. The peak at 1465 cm⁻¹ belongs to the bending vibration of –CH₂ and –CH₃; the peak at 1060 cm⁻¹ corresponds to the stretching of C–O while the peak at 720 cm⁻¹ is ascribed to the in-plane rocking vibration of –CH₂.²⁴ There was no peak for rGOAB

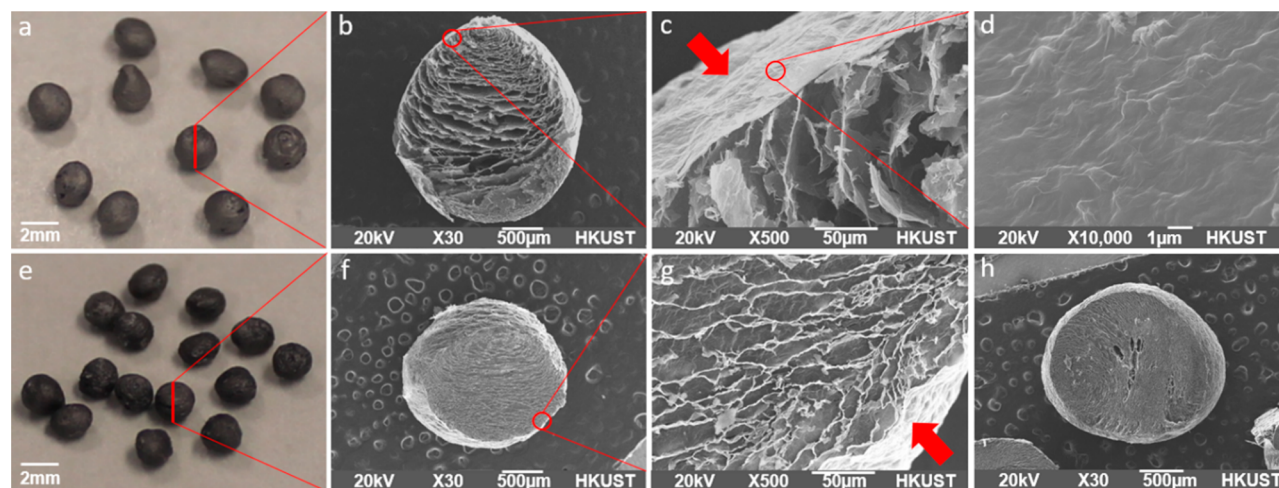


Figure 2. Macroscopic images of (a) neat and (e) TD-filled rGOABs. Cross-sectional SEM images of (b, c) neat and (f, g) TD-filled rGOAB, (d) rGOAB surface, and (h) TD-filled rGOAB after 50 cycles of heating and cooling.

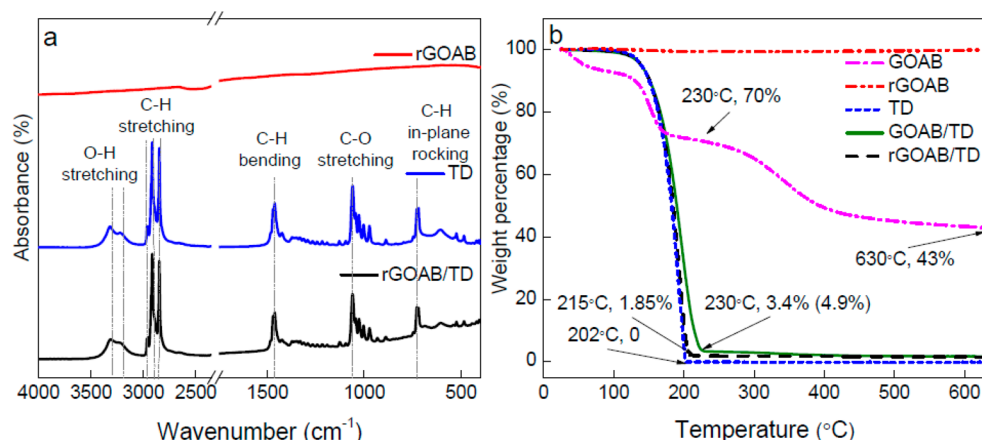


Figure 3. (a) FT-IR spectra for rGOAB, TD, and rGOAB/TD composite. (b) TGA curves of GOAB, rGOAB, TD, GOAB/TD, and rGOAB/TD of S15 and rS15 from room temperature to 630 $^{\circ}\text{C}$ in a nitrogen atmosphere.

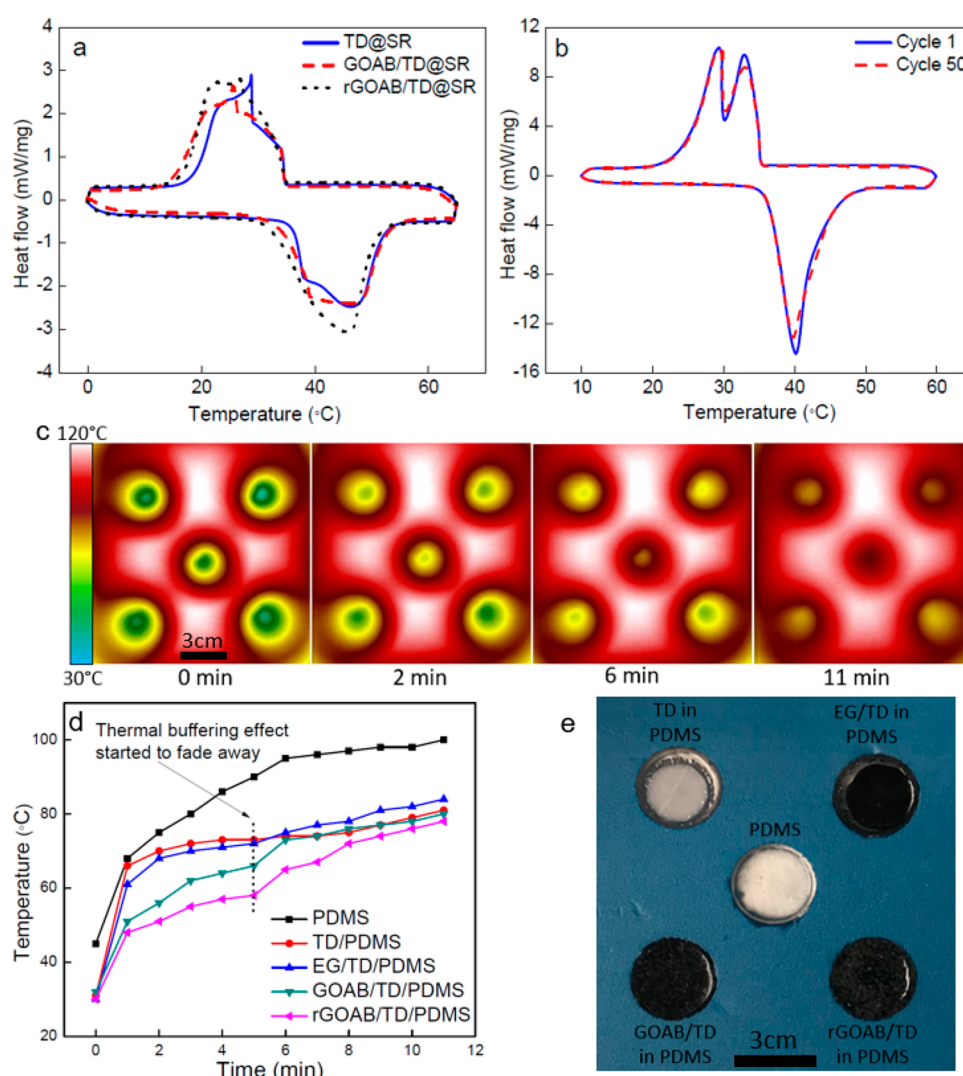


Figure 4. (a) Thermal responsive test results of three bead samples covered by silicone rubber at a heating rate of 2 $^{\circ}\text{C}/\text{min}$ and (b) thermal cyclic test results of rGOAB/TD powder at a heating rate of 20 $^{\circ}\text{C}/\text{min}$ on a DSC. (c) Infrared thermal images of PDMS, TD, EG/TD, GOAB/TD, and rGOAB/TD in PDMS placed on a hot plate at a temperature of 100 $^{\circ}\text{C}$. (d) Surface temperature variations with time during heating. (e) Digital images of the above-mentioned samples.

and the spectrum of rGOAB/TD, was much the same as that of TD with no new peak, signifying the encapsulation of TD in

rGOAB mainly by physical absorption without chemical interactions.

The absorbability of rGOAB and GOAB was further confirmed by TGA, and the results are shown in Figure 3b. The paraffin TD has vaporized at ~ 200 °C; therefore, the absorbability is defined as the percentage of paraffin impregnated in GOABs or rGOABs at this temperature. No obvious weight change was observed for rGOABs over the whole temperature range up to 630 °C studied, indicating that rGOABs were thermally stable and the weight loss of the rGOAB/TD composite arose solely from paraffin loss. At 215 °C, the curve of rGOAB/TD composite dropped to 1.85%, indicating 98.15% of paraffin weight fraction. Unlike rGOAB, the GOAB curve experienced a significant and gradual weight loss because of the presence of oxygenated functional groups in GO sheets which became reduced at high temperatures. Therefore, the weight loss of the sample GOAB/TD composite consisted of two parts: one arose from the evaporation of paraffin, and the other came from the reduction of oxygenated functional groups. At 230 °C, 70% of the weight of GOABs remained. At the very same temperature, the weight loss of the GOAB/TD composite was 96.6%. After excluding the reduction loss, the total paraffin weight fraction of GOAB/TD composite was 95.1%. This observation revealed that after thermal reduction the rGOABs showed higher absorbability of paraffin than GOABs, which verified the results from weighing. The stability test results performed at different temperatures, shown in Figure S1, indicated that both the composites containing over 95 wt % liquid-state paraffin TD remained quite stable, and no leakage was observed even after 2 h due to the capillary forces of the porous graphene networks.

The DSC curves in Figure 4a and the characteristics in Table S2 show the thermal responses of pure TD, GOAB/TD, and rGOAB/TD composites during phase transitions. TD melted into a liquid state above the phase change temperature and adhered to the bottom of the DSC pan, while the two composites were able to firmly hold paraffin TD within maintaining a small contact area with the pan bottom, which had a major effect on the DSC results. To make a fair comparison between the thermal responses of materials, instead of structures, all three materials with a comparable weight and shape were coated with a layer of silicon rubber having a thermal conductivity of ~ 0.4 W/(m K) to maintain comparable shapes. The phase transition of the rGOAB/TD and GOAB/TD composites came earlier than that of pure TD during the melting process, i.e., 33.7, 30.3, and 27.5 °C for TD, GOAB/TD, and rGOAB/TD, respectively, although the starting points of crystallization for all three material were similarly ~ 35 °C. This observation indicates that the structures of these materials, instead of crystallization behaviors, were responsible for different starting points of melting. Only the curve for rGOAB exhibited two peaks of the TD due to solid–solid phase transition,^{26,29,45,46} while these two peaks merged into one for the other materials, which is attributed to the higher thermal conductivity of rGO sheets and the micro-porous structure of rGOAB.

The DSC results after 50 heating–cooling cycles are shown in Figure 4b and Table S3. The almost identical two curves obtained from the 1st and 50th cycles mean high repeatability of heating and cooling cycles of the rGOAB/TD composite. It can be seen that the DSC curve of the rGOAB/TD after 50 heating–cooling cycles is very close to that of the rGOAB/TD at the very first cycle of the heating–cooling test. The average latent heat of 222.5 J g^{−1} that remained after 50 cycles is equivalent to above 96.6% of the initial value, 230.3 J g^{−1}. Only

the melting temperature and the liquid-disordered solid phase change temperature marginally downshifted while there were no shifts for the disordered and ordered solid phase change temperatures, signifying excellent thermal reversibility and cyclic stability of the rGOAB/TD composite. Interestingly, the latent heat value for rGOAB/TD, 230.3 J g^{−1}, did not change much compared with pure TD, the latent heat of which was 230.4 J g^{−1}. This can be explained by the fact that even though the TD concentration was compromised by adding rGOABs, the total latent heat was enhanced by better crystallinity of organic PCM in the presence of spongy graphene sheets as previously reported.²⁹

The isothermal profiles and the temperature changes of five different materials on a hot plate maintained a constant temperature of 130 °C are shown in Figure 4c along with their positions in Figure 4e. It can be seen in both Figures 4c and 4d that the temperature of PDMS rapidly increased without any thermal buffer, while other four paraffin TD-containing materials showed thermal buffering with lower temperature. The temperatures of GOAB/TD and rGOAB/TD in PDMS materials were lower than those of TD and EG/TD PDMS materials, which can be explained by the more heat pathway provided by well-distributed beads with higher surface area than bulky TD and EG/TD. Paraffin TD absorbed heat faster when encapsulated in beads and uniformly distributed in a matrix than those in pure bulky form or encapsulated in EG. From Figure 4d, it can be seen that the thermal buffering effect started to fade away after 5 min, with prominent temperature increases for EG/TD, GOAB/TD, and rGOAB/TD materials. The case for TD in PDMS material was different, with no obvious temperature increase observed around 5 min or later, which means that it took longer for pure paraffin TD to be totally melted. This can be explained by the low heat absorbing rate of bulky PCM due to the lower thermal conductivity of pure paraffin TD over EG/TD and low surface area between bulky TD and the matrix compared with GOAB/TD and rGOAB/TD. The temperature of the EG/TD material was lower than pure TD for the first 5 min, indicating faster thermal response due to the higher thermal conductivity of EG/TD.^{8,12,16} Between GOAB/TD and rGOAB/TD materials, the temperature of GOAB/TD was higher than that of rGOAB/TD, indicating better thermal responsive performance of rGOAB/TD due to higher heat transfer property after thermal annealing. The thermal conductivities of these samples are shown in Figure S2. It is shown that they were not constant, but rather fluctuated within a small range without obvious pattern. It is suspected that the overall thermal conductivities of PDMS-based composites did not contribute to the difference in thermal response of different materials. Instead, the local thermal conductivity and local temperature distribution dominated the thermal response tests. Figures S3 and S4 show the cooling counterpart of the test, demonstrating difference in thermal buffering among different materials. In summary, uniformly distributed beads loaded with PCM show better thermal response than bulky PCM of the same weight, with rGOABs exhibiting faster response than GOABs, which demonstrates the advantages of the rGOABs. The Young's modulus of EG/TD and rGOAB/TD/SR samples at room temperature (RT) and 40 °C were obtained from a compression test conducted at both temperatures as shown in Figure S5. The results showed that above the phase change temperature of TD the mechanical strength of EG/TD deteriorated to a large extent, while rGOAB/TD/SR

maintained the strength with a small decrease in Young's modulus.

To verify the benefits of the rGOAB/TD composites for PCM encapsulation, comprehensive comparisons were performed between rGOAB/TD materials and other TD-encapsulated composites with various supporting materials and methods, as shown in Figure 5. The details of their

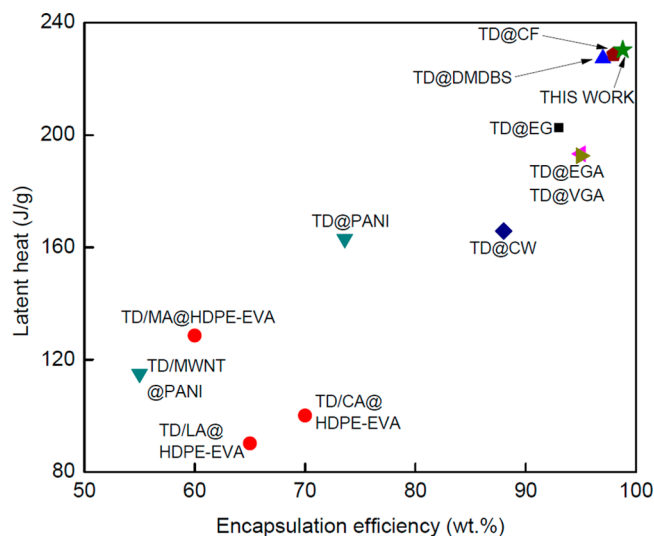


Figure 5. Comparison of latent heat as a function of encapsulation efficiency of materials: TD in EG,⁴⁷ TD/capric acid (CA) in high-density polyethylene–ethylene–vinyl acetate (HDPE-EVA),⁴⁸ TD/lauric acid (LA) in HDPE-EVA,⁴⁸ TD/myristic acid (MA) in HDPE-EVA,⁴⁸ TD in 1,3:2,4-di(3,4-dimethyl)benzylidene sorbitol (DMDBS),⁴⁹ TD/MWNT in polyaniline (PANI),⁵⁰ TD in PANI,⁵¹ TD in ethylenediamine–graphene aerogel (EGA),²⁴ TD in vitamin C–graphene aerogel (VGA),²⁴ TD in carbonized wood (CW),⁵² and beeswax-TD in carbon fiber (CF)⁵³ and this work.

encapsulation and thermal properties are given in Table S4. The melting temperatures of the TD encapsulated materials studied varied from 21.7 to 40.2 °C, mostly focusing on ~35 °C. The encapsulation efficiencies range from ~55 to ~98 wt %. The highest efficiencies reported were TD encapsulated in carbon fiber (CF)⁵³ with an encapsulation efficiency of 98.0 wt % and latent heat of 228.7 and 227.6 J g⁻¹ for melting and freezing, respectively, followed by TD in DMDBS⁴⁹ with an efficiency of 97.0 wt % and latent heat of 227.3 and 226.7 J g⁻¹. The rGOAB/TD composite in this study presented an extremely high encapsulation efficiency of 98.3 wt % with high latent heat of 232.2 and 228.4 J g⁻¹ at a melting temperature of 40 °C, which makes it an ideal thermal management material, especially for applications where high energy density is desired and operating temperature is below 40 °C, e.g., lithium-ion battery pack.

4. CONCLUSIONS

A novel form-stable rGOAB/TD composite was fabricated by the wet-spinning and impregnation method. For the composite, TD is used as the thermal storage material, and rGOAB acts as the supporting material as well as improving the applicational flexibility of the composite. The latent heat of the composite was determined to be 230.3 J g⁻¹ by DSC and remain above 96.6% after 50 cycles of melting and freezing. The composition characteristics of rGOAB/TD composite are close to those of the pure TD and GO because there is no

chemical reaction between the TD and hollow rGOAB in preparation of the composite. The rGOAB/TD composite contained up to 98.83 wt % of TD, which is the highest mass fraction ever reported. rGOABs loaded with TD shows faster thermal response over EG/TD and GOAB/TD to prove the excellent heat transfer properties of rGOAB and the advantage of separated encapsulated beads over bulky materials. In conclusion, the as-prepared composite has a high heat storage capability, good thermal stability, fast thermal response, and high flexibility, indicating that the composite can be used as a potential material to store thermal energy in various thermal management applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsam.9b00374.

Raw materials and processing parameters; form-stability test; DSC characteristics; thermal conductivity results; infrared thermal images and temperature variation for cooling process; Young's modulus of EG/TD and rGOAB/TD@SR at different temperatures; energy storage properties of TD-based composites reported in the literature (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The work was financially supported by the Hong Kong University of Science and Technology (Grant#: R9365) of Hong Kong SAR and Guangdong Science and Technology Department (Project#: 2017A050506005 and 2018B050502001).

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